Chapter 18, Problem 68CP

Problem

The voltage signal at the input of a filter is $v(t) = 50e{-2|t|}$ V. What percentage of the total 1-**Ω** energy content lies in the frequency range of $1<\omega<5$ rad/s?

Step-by-step solution

Step 1 of 1

The total energy is

$$W_T = \int_{-\infty}^{\infty} V^2(t)\,dt$$

since $\mathrm{V(t)}$ is an even function

$$W_T = \int_0^{\infty} 2500e^{-4t}\,dt = 5000\left.\frac{e^{-4t}}{-4}\right|_0^{\infty} = 1250J$$

$$V(\omega) = \frac{50\times 4}{4+\omega^2}$$

$$W = \frac{1}{2\pi}\int_1^5 1V(\omega)^2\,|^2\,d\omega = \frac{1}{2\pi}\int_1^5 \frac{(200)^2}{\left(4+\omega^2\right)^2}d\omega$$

$$but \quad \int \frac{1}{\left(a^2+x^2\right)^2} = \frac{1}{2a^2}\left[\frac{x}{x^2+a^2}+\frac{1}{a}+an^{-1}\left(\frac{x}{a}\right)\right]+c$$

$$W = \frac{2\times 10^4}{\pi}\frac{1}{8}\left[\frac{\omega}{4+\omega^2}+\frac{1}{2}\tan^{-1}\frac{\omega}{2}\right]_1^5$$

$$= \left(\frac{2500}{\pi}\right)\left(\frac{5}{29}\right)+0.5\tan^{-1}\left(\frac{5}{2}\right)-\left(\frac{1}{5}\right)-0.5\tan^{-1}\left(\frac{1}{2}\right)$$

$$\frac{W}{W^T} = \frac{267.19}{1250} = 0.2137 = 21.37\%$$